Method Note | Data Pipeline, Formulas & Reproducibility

BRICKHUB INTELLIGENCE | Data Governance, Verification Protocols & Technical Specifications

HYBRID DECISION PROTOTYPE — Verified Public Infrastructure + Calibrated Market Deed Records

1. End-to-End Pipeline

Raw Public Records (IGR, MahaRERA, Rental, Transit Authorities) → Ingestion & Schema Validation → 7 Derived Indicator Calculations → Missing Data Imputation → Min-Max Normalization (0–100) → Multi-Criteria Weighted Sum → Monte Carlo & Leave-One-Out Stress Testing → Final Micro-Market Ranking.

2. Mathematical Formulations

- YoY Price Growth (%) = 100 × (Median Rate Current 12M / Median Rate Prior 12M – 1)
- Sales-to-Launch Ratio = Units Sold 12M / Units Launched 12M
- Gross Rental Yield (%) = 100 × (Monthly Rent × 12) / (Median Rate ■/sqft × Representative Area sqft)
- Months of Inventory = 12 × Unsold Inventory / Annual Units Sold (Inverted: lower overhang gets higher score)
- Transaction Liquidity Score = 70% × min(100 × Txn12m / 3000, 100) + 30% × clip(50 + 100 × (Txn12m / Txn_prev – 1), 0, 100)
- Risk & Affordability Score = 70% × (100 – Risk Score) + 30% × clip(50 – 1.5 × Premium To Peer %, 0, 100)

3. Normalization & Missing Data Mechanics

Each metric is normalized across eligible markets: Score_norm = 100 × (x – min) / (max – min). For unfavorable metrics (inventory overhang), score is inverted: 100 × (max – x) / (max – min). Constant metrics default to 50. Three missing-data modes are implemented: Median imputation (baseline), Conservative (10th percentile for positives, 90th for negatives), and Complete-case (drops incomplete markets).

4. Verification & QA Checks

Automated unit testing via pytest checks: deterministic reproducibility (seed 42), exact mathematical equations, boundary checks (negative prices, zero denominators), non-negative weights summing to 100%, and risk inversion validation (higher risk always strictly decreases composite score).

5. AI Tools Disclosure

Per assignment instructions: AI assistants and web search tools were utilized to extract public infrastructure progress reports (PMRDA Line 3 and Maha Metro updates), format boilerplate Streamlit UI code, structure unit tests, and compile long-format provenance tables. All indicator selection, weights, scoring logic, and sensitivity analyses were human-supervised.